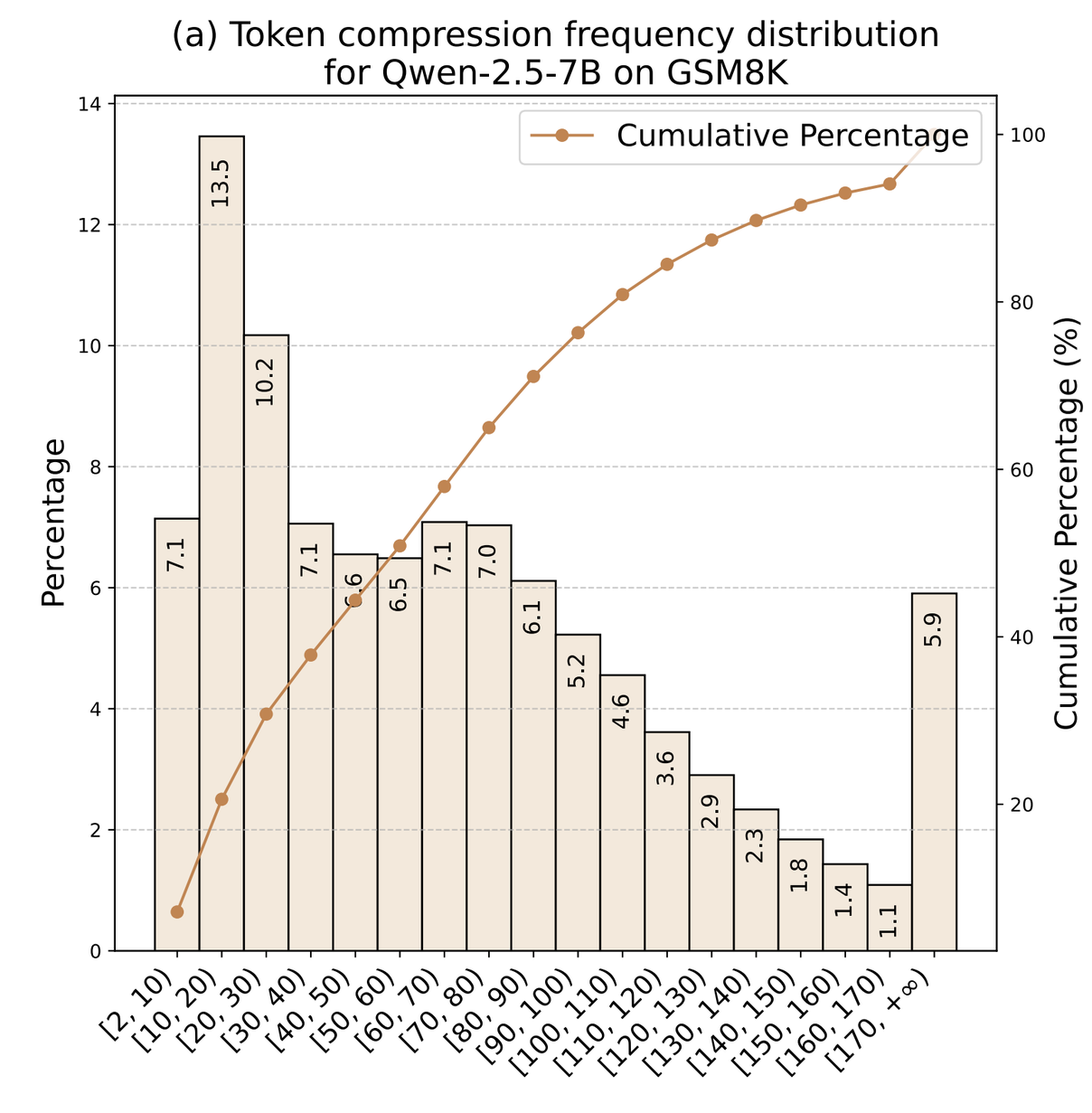
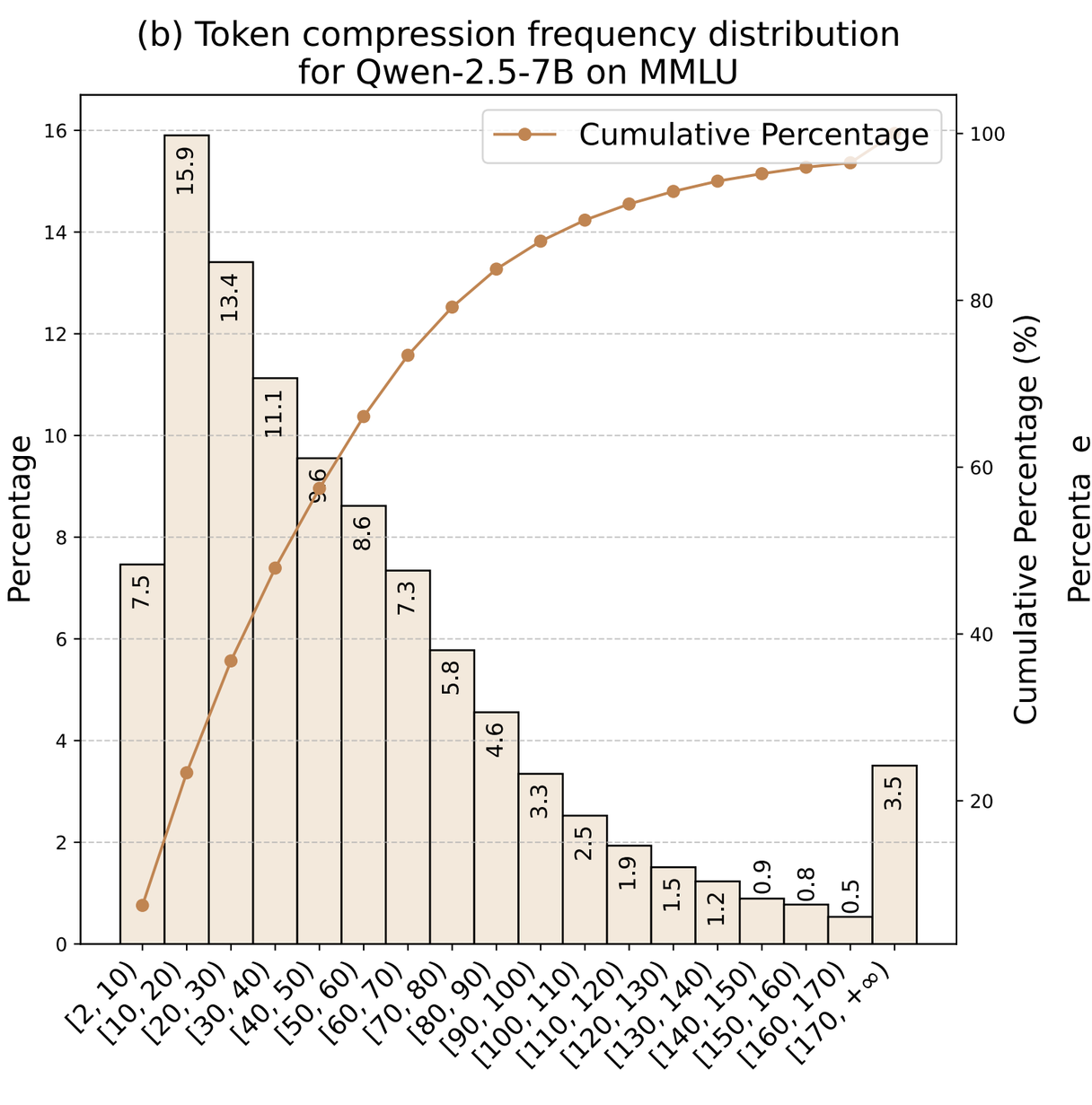
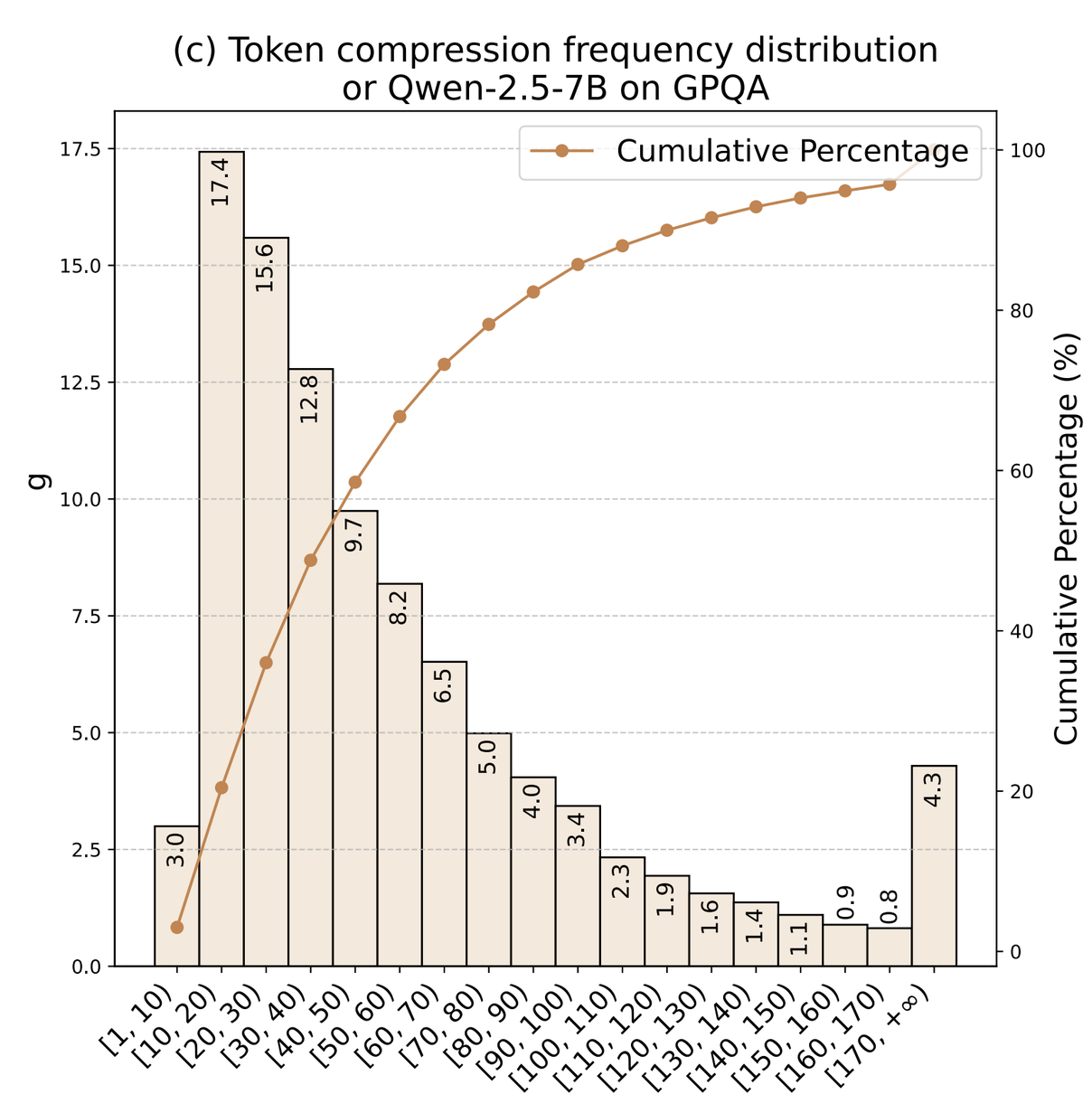
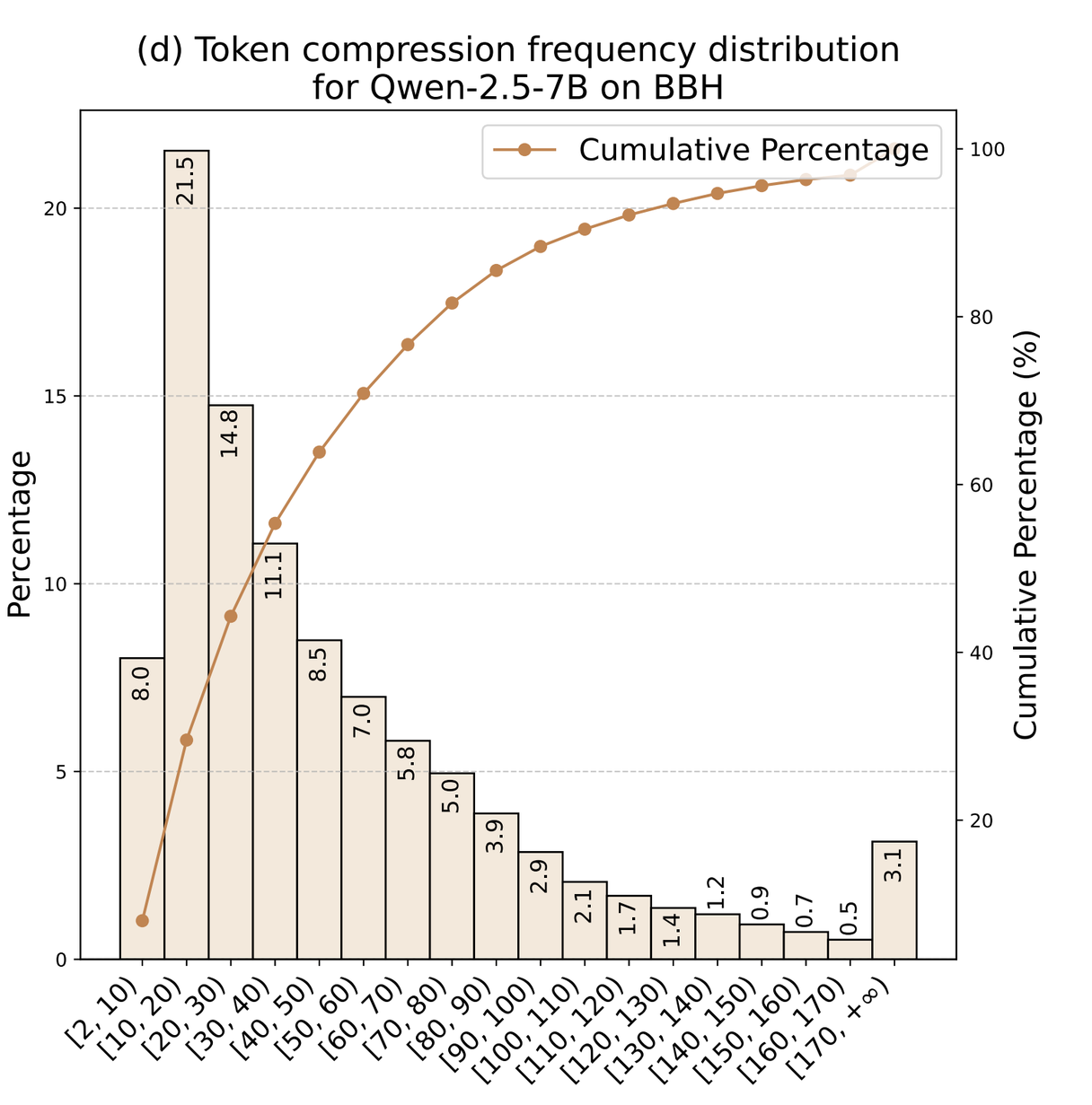
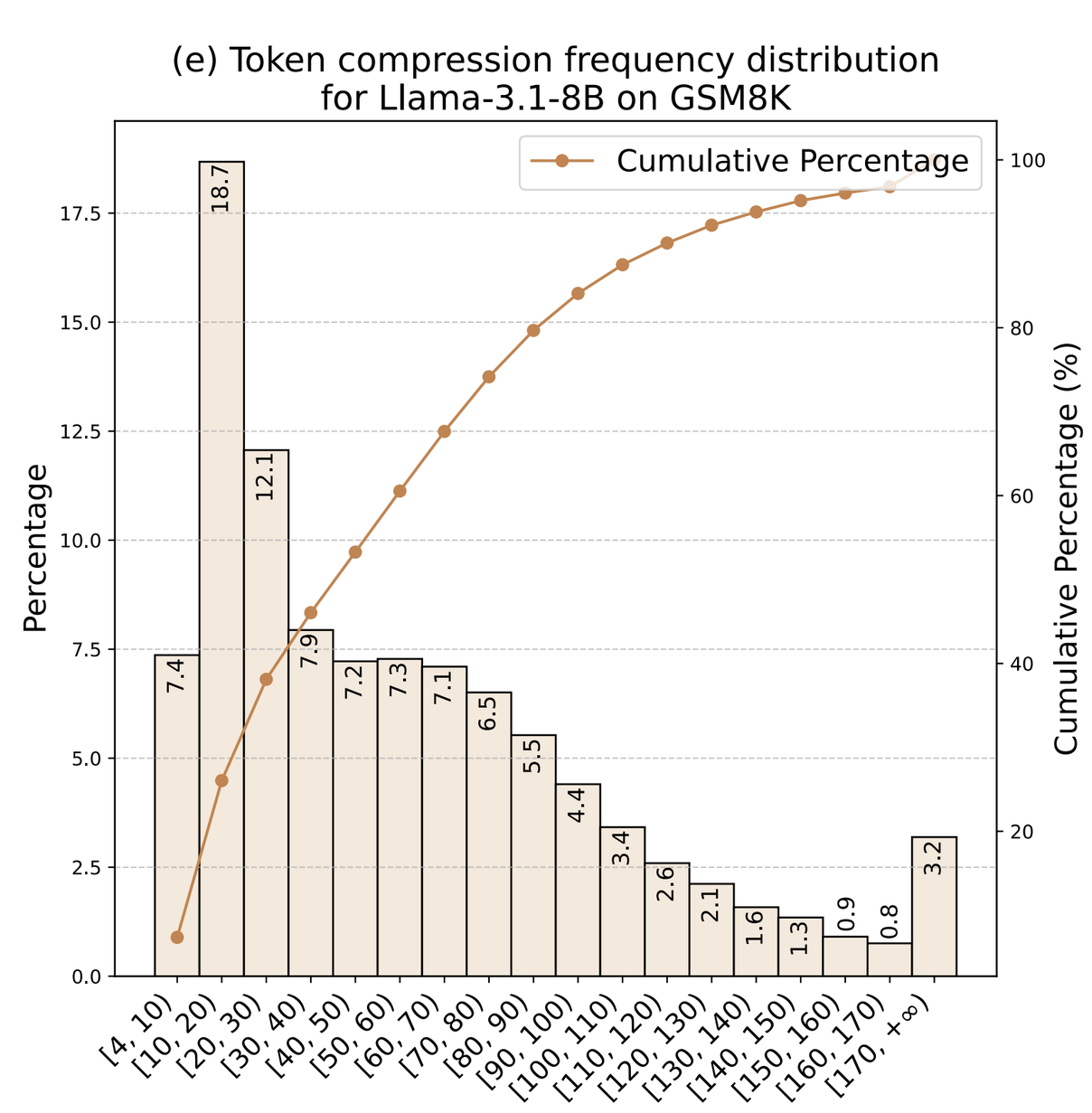
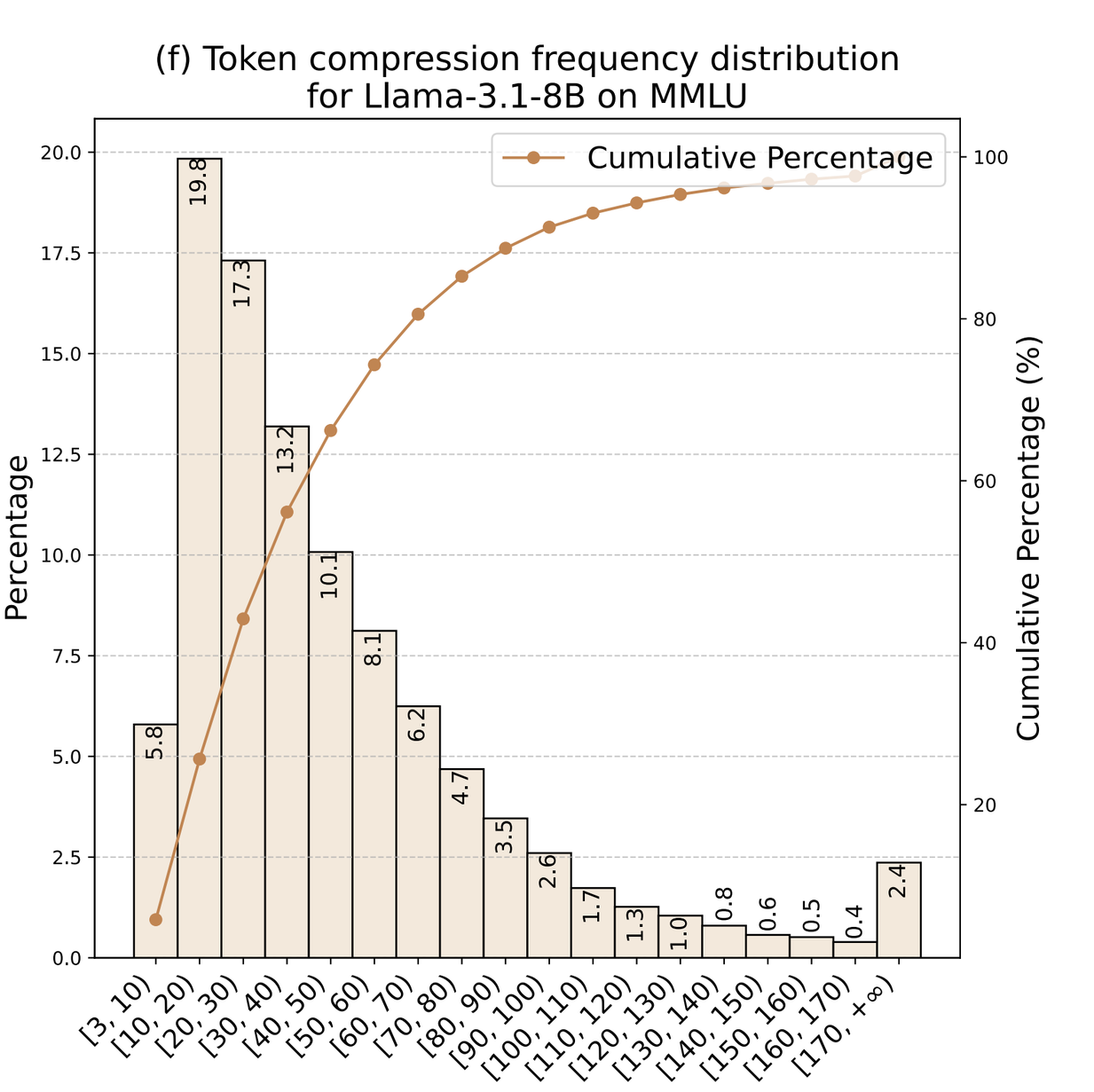

(a) Token compression frequency distribution
for Qwen-2.5-7B on GSM8K
Cumulative Percentage
Percentage
Cumulative Percentage (%)
7.1
13.5
10.2
7.1
6.6
6.5
7.1
7.0
6.1
5.2
4.6
3.6
2.9
2.3
1.8
1.4
1.1
5.9
[2, 10)
[10, 20)
[20, 30)
[30, 40)
[40, 50)
[50, 60)
[60, 70)
[70, 80)
[80, 90)
[90, 100)
[100, 110)
[110, 120)
[120, 130)
[130, 140)
[140, 150)
[150, 160)
[160, 170)
[170, +∞)

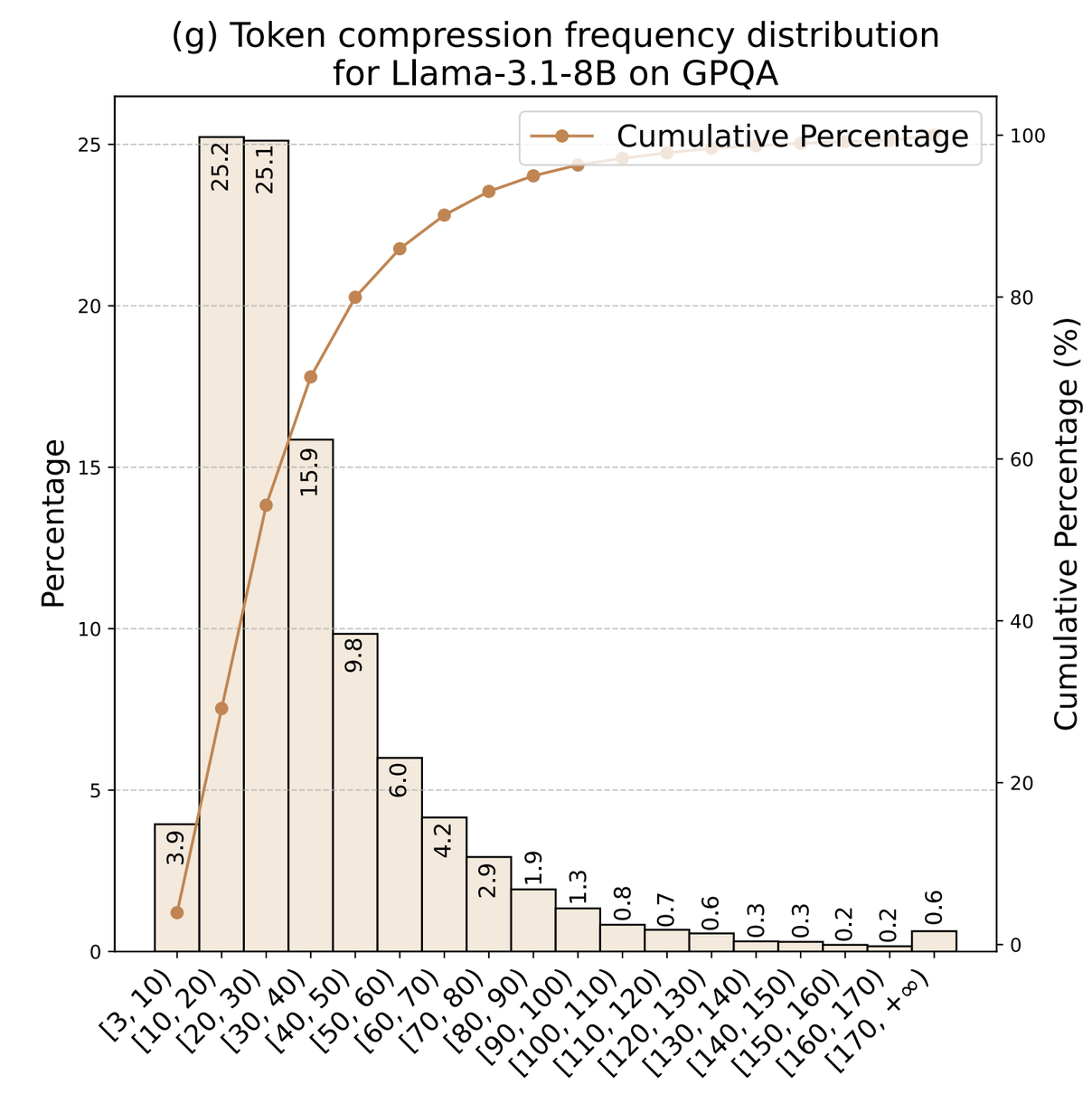

(b) Token compression frequency distribution
for Qwen-2.5-7B on MMLU
Cumulative Percentage
Percentage
Cumulative Percentage (%)
7.5
15.9
13.4
11.1
9.6
8.6
7.3
5.8
4.6
3.3
2.5
1.9
1.5
1.2
0.9
0.8
0.5
3.5
[2, 10)
[10, 20)
[20, 30)
[30, 40)
[40, 50)
[50, 60)
[60, 70)
[70, 80)
[80, 90)
[90, 100)
[100, 110)
[110, 120)
[120, 130)
[130, 140)
[140, 150)
[150, 160)
[160, 170)
[170, +∞)

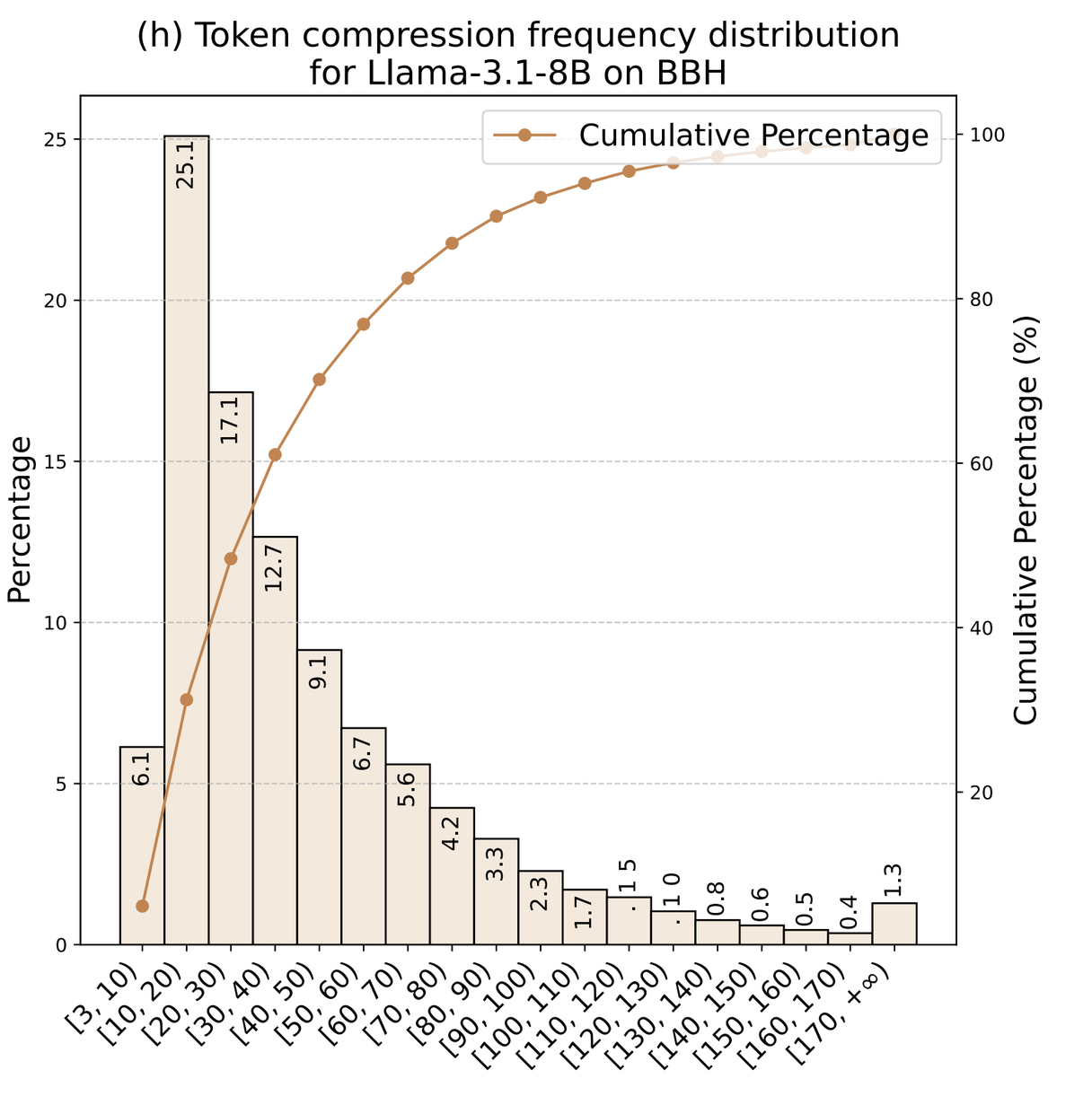

(c) Token compression frequency distribution
or Qwen-2.5-7B on GPQA
Cumulative Percentage
g
Cumulative Percentage (%)
3.0
17.4
15.6
12.8
9.7
8.2
6.5
5.0
4.0
3.4
2.3
1.9
1.6
1.4
1.1
0.9
0.8
4.3
[1, 10)
[10, 20)
[20, 30)
[30, 40)
[40, 50)
[50, 60)
[60, 70)
[70, 80)
[80, 90)
[90, 100)
[100, 110)
[110, 120)
[120, 130)
[130, 140)
[140, 150)
[150, 160)
[160, 170)
[170, +∞)


(d) Token compression frequency distribution
for Qwen-2.5-7B on BBH
Cumulative Percentage
Percentage
Cumulative Percentage (%)
8.0
21.5
14.8
11.1
8.5
7.0
5.8
5.0
3.9
2.9
2.1
1.7
1.4
1.2
0.9
0.7
0.5
3.1
[2, 10)
[10, 20)
[20, 30)
[30, 40)
[40, 50)
[50, 60)
[60, 70)
[70, 80)
[80, 90)
[90, 100)
[100, 110)
[110, 120)
[120, 130)
[130, 140)
[140, 150)
[150, 160)
[160, 170)
[170, +∞)


(e) Token compression frequency distribution
for Llama-3.1-8B on GSM8K
Cumulative Percentage
Percentage
Cumulative Percentage (%)
7.4
18.7
12.1
7.9
7.2
7.3
7.1
6.5
5.5
4.4
3.4
2.6
2.1
1.6
1.3
0.9
0.8
3.2
[4, 10)
[10, 20)
[20, 30)
[30, 40)
[40, 50)
[50, 60)
[60, 70)
[70, 80)
[80, 90)
[90, 100)
[100, 110)
[110, 120)
[120, 130)
[130, 140)
[140, 150)
[150, 160)
[160, 170)
[170, +∞)


(f) Token compression frequency distribution
for Llama-3.1-8B on MMLU
Cumulative Percentage
Percentage
Cumulative Percentage (%)
5.8
19.8
17.3
13.2
10.1
8.1
6.2
4.7
3.5
2.6
1.7
1.3
1.0
0.8
0.6
0.5
0.4
2.4
[3, 10)
[10, 20)
[20, 30)
[30, 40)
[40, 50)
[50, 60)
[60, 70)
[70, 80)
[80, 90)
[90, 100)
[100, 110)
[110, 120)
[120, 130)
[130, 140)
[140, 150)
[150, 160)
[160, 170)
[170, +∞)


(g) Token compression frequency distribution
for Llama-3.1-8B on GPQA
Cumulative Percentage
Percentage
Cumulative Percentage (%)
3.9
25.2
25.1
15.9
9.8
6.0
4.2
2.9
1.9
1.3
0.8
0.7
0.6
0.3
0.3
0.2
0.2
0.6
[3, 10)
[10, 20)
[20, 30)
[30, 40)
[40, 50)
[50, 60)
[60, 70)
[70, 80)
[80, 90)
[90, 100)
[100, 110)
[110, 120)
[120, 130)
[130, 140)
[140, 150)
[150, 160)
[160, 170)
[170, +∞)


(h) Token compression frequency distribution
for Llama-3.1-8B on BBH
Cumulative Percentage
Percentage
Cumulative Percentage (%)
6.1
25.1
17.1
12.7
9.1
6.7
5.6
4.2
3.3
2.3
1.7
1.5
1.0
0.8
0.6
0.5
0.4
1.3
[3, 10)
[10, 20)
[20, 30)
[30, 40)
[40, 50)
[50, 60)
[60, 70)
[70, 80)
[80, 90)
[90, 100)
[100, 110)
[110, 120)
[120, 130)
[130, 140)
[140, 150)
[150, 160)
[160, 170)
[170, +∞)